Browser Object Model (BOM)

- The Browser Object Model (BOM) allows JavaScript to interact with the web browser. It provides objects that represent the browser window and browser-related information.

- BOM is used to working with the browser window.
- It provides information about the current URL, browser, screen, and browsing history.
- The window object is the main object of the BOM.

BOM Structure

window
- location
- history
- navigator
- screen

The window object represents the browser window. Other browser-related objects can be accessed through window.

1. Window Object

The window object represents the browser window in which the web page is displayed. It is the top-level object available in browser-side JavaScript.

```
window.alert("Welcome");
```

In many cases, window can be omitted.

```
alert("Welcome");
```

Common Window Properties

Property	Purpose
innerWidth	Returns the width of the browser's content area.
innerHeight	Returns the height of the browser's content area.

```
console.log(window.innerWidth);
console.log(window.innerHeight);
```

Common Window Methods

Method	Purpose
alert()	Displays a message box.
confirm()	Displays OK and Cancel buttons.
prompt()	Accepts input from the user.
open()	Opens a new browser window or tab.
close()	Attempts to close a window opened by script.

alert()

The alert() method displays a message to the user.

```
alert("Welcome to JavaScript");
```

confirm()

The confirm() method displays a dialog box with OK and Cancel buttons. It returns true when OK is selected and false when Cancel is selected.

```
let result = confirm("Do you want to continue?");
console.log(result);
```

prompt()

The prompt() method displays a dialog box that allows the user to enter a value.

```
let name = prompt("Enter your name");
console.log(name);
```

2. Location Object

The location object contains information about the current page URL and can be used to navigate to another page.

console.log(window.location.href);

Common Location Properties

Property	Purpose
href	Returns the complete URL.
hostname	Returns the domain name.
pathname	Returns the path of the current page.
protocol	Returns the URL protocol.

Common Location Methods

Method	Purpose
assign()	Loads another page.
reload()	Reloads the current page.
replace()	Loads another page and replaces the current history entry.

location.assign("https://example.com");
location.reload();

3. History Object

It represents the pages visited in the current browser session. It can be used to move backward or forward through the browsing history.

Common History Methods

Method	Purpose
back()	Moves to the previous page.
forward()	Moves to the next page.
go()	Moves to a particular position in the session history.

history.back();
history.forward();
history.go(-1);

history.go(-1) moves one page backward in the session history.

4. Navigator Object

The navigator object provides information about the browser and its environment.

Common Navigator Properties

Property	Purpose
userAgent	Returns information about the browser's user-agent string.
language	Returns the browser's preferred language.
onLine	Indicates whether the browser reports that it is online.

```
console.log(navigator.userAgent);
console.log(navigator.language);
console.log(navigator.onLine);
```

Note: Browser identification should not normally rely only on userAgent because its value can vary or be modified.

5. Screen Object

The screen object provides information about the user's screen.

Common Screen Properties

Property	Purpose
width	Returns the screen width in CSS pixels.
height	Returns the screen height in CSS pixels.
availWidth	Returns the available screen width.
availHeight	Returns the available screen height.
colorDepth	Returns the color depth of the screen.

```
console.log(screen.width);
console.log(screen.height);
```